

Maria Fortunato

📍 Porto - Portugal ✉ mariafortunatom@gmail.com ☎ +351 912 068 769 in mariafortunatom 🌐 mfortuna23

Software Developer

I'm a software developer with hands-on experience building systems-level projects in C and C++. I enjoy the logical challenge of solving complex problems, whether it's optimizing algorithms or implementing multithreading solutions. Through projects like writing a UNIX shell and solving concurrency challenges, I've developed strong problem-solving skills and learned to work effectively in team environments. I'm now looking for a junior position where I can continue growing my skills while contributing to meaningful projects.

Projects

IRC Team Project (C++, Network Programming)

- Built IRC server supporting multiple concurrent clients using socket programming
- Implemented RFC 1459 protocol with channels, private messaging, and operator commands
- Used I/O multiplexing (poll) for efficient connection handling without blocking

Cub3D Team Project (C, Raycasting)

- Developed a performant 3D raycasting engine from scratch, implementing textured walls and colored floor/ceiling.
- Implemented WASD movement, camera rotation with arrows and designed efficient wall collision detection.

Minishell Team Project (C, UNIX)

- Developed a Bash-like shell with pipe (|) and redirection (>, <) support;
- Created a robust lexer to handle complex command combinations with pipes and redirections.
- Implemented advanced shell features including environment variable expansion, signal handling, and proper quoting/escaping rules while preventing memory leaks.

Philosophers (C, Multithreading)

- Solved the classic concurrency problem by implementing a thread-safe solution using mutexes and careful resource timing to prevent deadlocks. Designed a monitoring system to track each philosopher's state (eating/ thinking/ sleeping) while avoiding starvation.

Education

42 Porto , Computer Science

Aug 2023 – Present

- Project-based learning in algorithms, systems programming, and peer collaboration
- Developed systems-level projects in C/C++, emphasizing efficiency, debugging, and workflow optimization.
- Peer-to-peer learning methodology honed collaboration, adaptability, and autonomous project management.

Instituto Superior de Serviço Social do Porto , Social Work

Set 2016 – Apr 2020

Work Experience

Livraria Lello, Customer Assistant

Jun 2024 – Present

Viral VR, Customer Assistant

Jan 2024 – Jun 2024

Mais Família Lda., Social Worker

Apr 2022 – Jul 2023

QPI - Qualificar Para incluir, Social Worker

Jan 2021 – Jul 2021

Skills

Technical skills: C and C++ | Memory management | Multithreading | UNIX Shell | Git version control

Soft skills: Problem-Solving | Team Collaboration | Adaptability | Communication

Languages: Portuguese (Native) | English (Advanced) | Spanish (Intermediate)

Certificates: Data Analyst SQL (CESAE digital)